

Tiptop Virtual Academy Ontology

The Canonical Model of the Academy

Version: 2.0 Foundation Draft

Purpose

The TVA Ontology defines the fundamental concepts that exist within Tiptop Virtual Academy and the relationships between them.

It provides a common language for educators, administrators, developers, AI systems, and future stakeholders.

The ontology is the single source of truth for understanding the Academy.

Every policy, workflow, report, AI agent, database, API, interface, and business process shall derive its meaning from this document.

Ontological Principles

Every entity within the Academy must answer six questions:

1. What is it?

Its definition.

2. Why does it exist?

Its purpose.

3. Who owns it?

Its business owner.

4. Who interacts with it?

People or systems.

5. Where does it live?

Its authoritative domain.

6. What relationships does it have?

How it connects to other entities.

Level One — Human Entities

These are people who participate in the Academy.

Applicant

A person who has requested admission but has not yet been accepted.

Exists to begin the admissions journey.

Transitions into:

Accepted Applicant

or

Rejected Applicant

Student

An applicant who has completed enrollment and officially belongs to the Academy.

Purpose:

To participate in educational programmes.

Owning Domain:

Student Services

Relationships:

- belongs to Parents
 - enrolled in Programmes
 - studies Subjects
 - attends Lessons
 - completes Assessments
 - receives Reports
 - owns Academic Records
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Parent / Guardian

The individual legally or personally responsible for one or more students.

Purpose:

Educational partnership.

May be linked to multiple students.

Teacher

A qualified educator responsible for facilitating learning.

Owns:

Lesson delivery.

Assessment.

Academic guidance.

May teach many subjects.

Staff Member

An employee who supports institutional operations outside direct teaching responsibilities.

Examples:

Finance

Admissions

HR

Support

Marketing

Administration

Level Two — Institutional Entities

Programme

A complete educational offering.

Examples:

Primary Education

Secondary Education

Holiday Programme

One-to-One Tutoring

A Programme contains Subjects.

Subject

A structured field of study.

Examples:

English

Mathematics

Physics

Biology

History

A Subject belongs to one or more Programmes.

Contains Lessons.

Contains Assessments.

Academic Session

The highest academic time boundary.

Example:

2026/2027

Contains Terms.

Academic Term

A subdivision of an Academic Session.

Contains Weeks.

Contains Teaching Activities.

Class Group

A community of learners progressing together through a programme during a defined academic period.

Examples:

Year 5A

Primary 3

IGCSE Biology Cohort

A Class Group is **not** the same as a virtual classroom or a video meeting. It represents the academic cohort.

Enrollment

The formal relationship that connects a student to a programme for a specific academic session.

Enrollment has a lifecycle (pending, active, completed, withdrawn, transferred) and is the authoritative record of a student's participation.

Level Three — Learning Entities

Lesson

A planned learning experience with defined objectives and outcomes.

A Lesson belongs to a Subject and may be delivered through different methods (live, recorded, blended, independent).

Learning Activity

Any structured task designed to promote learning.

Examples:

Reading

Discussion

Quiz

Experiment

Project

Reflection

A Lesson may contain multiple Learning Activities.

Assignment

A Learning Activity that requires learner completion and submission.

Produces evidence of learning.

Assessment

A structured evaluation of learning.

May include:

Quiz

Test

Examination

Project

Presentation

Practical Work

Produces Grades and Feedback.

Grade

A formal evaluation resulting from an Assessment.

Belongs to a Student for a specific Assessment.

Feedback

Qualitative guidance intended to improve learning.

Feedback may accompany Grades but also exists independently.

Level Four — Operational Entities

Admission Application

The official request to join the Academy.

Invoice

A request for payment.

Created by Finance.

Paid by Parent.

Linked to Enrollment.

Payment

Evidence that an Invoice has been settled.

Attendance Record

Evidence of participation in a scheduled learning experience.

Communication

An official exchange of information.

May occur between:

School ↔ Parent

Teacher ↔ Student

Administration ↔ Staff

System ↔ User

Policy

A governing rule adopted by the Academy.

Policies influence workflows but do not replace professional judgment.

Level Five — Digital Entities

These entities exist because the Academy operates digitally, but they are subordinate to the institutional model.

User Account

The digital identity through which a person accesses Academy systems.

A User Account represents a person within technology.

It is not the person.

Role

A collection of permissions granted to a User Account.

Examples:

Student

Teacher

Parent

Administrator

Finance Officer

Permission

Authority to perform a defined action within Academy systems.

Permissions belong to Roles rather than directly to individuals wherever practical.

Notification

A system-generated message informing users of relevant events or required actions.

Digital Resource

Any electronic material used to support learning or operations.

Examples:

Documents

Videos

Presentations

Interactive Content

Recorded Lessons

Fundamental Relationships

The Academy recognizes relationships as first-class concepts.

Examples include:

Applicant → becomes → Student

Student → has → Enrollment

Enrollment → belongs to → Programme

Programme → contains → Subject

Subject → contains → Lesson

Lesson → contains → Learning Activities

Assignment → produces → Submission

Assessment → produces → Grade

Teacher → delivers → Lesson

Parent → supports → Student

Invoice → requests → Payment

Payment → activates → Enrollment (where applicable)

Canonical Truths

The following statements are always true within the Academy:

- A Student cannot exist without an Enrollment.
 - An Enrollment cannot exist without a Programme.
 - A Lesson cannot exist without a Subject.
 - A Subject may belong to multiple Programmes.
 - A Parent may support multiple Students.
 - A Teacher may teach multiple Subjects.
 - A User Account represents—but is never confused with—a person.
 - Every official record has one authoritative owner.
 - Every entity has a defined lifecycle.
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Closing Statement

The TVA Ontology is the language of the Academy.

It defines what exists, how those things relate, and how they evolve over time.

It is intentionally independent of curriculum, software platforms, database technologies, and organizational structures.

By establishing a shared understanding of reality, the ontology provides the stable foundation upon which every future policy, workflow, AI capability, software system, and educational innovation can confidently be built.